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RESEARCH ARTICLE

Using Machine Learning to Capture Heterogeneity in Trade Agreements

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Abstract

This paper uses machine learning techniques to capture heterogeneity in free trade agreements. The tools of machine learning allow us to quantify several features of trade agreements, including volume, comprehensiveness, and legal enforceability. Combining machine learning results with gravity analysis of trade, we find that more comprehensive agreements result in larger estimates of the impact of trade agreements. In addition, we identify the policy provisions that have the most substantial effect on creating trade flows. In particular, legally binding provisions on antidumping, capital mobility, competition, customs harmonization, dispute settlement mechanism, e-commerce, environment, export and import restrictions, freedom of transit, investment, investor-state dispute settlement, labor, public procurement, sanitary and phytosanitary measures, services, technical barriers to trade, telecommunications, and transparency tend to have the largest trade creation effects.

Keywords Free trade agreements · Machine learning · Gravity model · Deep integration

JEL Classification F10 · F13

1 Introduction

The gravity model—often referred to as the workhorse model in international trade—has been widely used to study the effects of various determinants of trade flows across countries. Drawing from the analogy of physical science, Tinbergen (1962) first used the gravity equation to evaluate the impact of free trade agreements (FTAs) on bilateral trade flows. Since Tinbergen (1962), numerous papers have studied

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the role of various determinants of trade flows, such as distance, adjacency, common language, presence of a bilateral agreement, and past colonial links, to name a few (cf. Head and Mayer 2014). One of the more widely studied factors in the empirical gravity literature is the impact of economic integration agreements (EIAs) on trade flows. This interest in the effects of EIAs on trade flows is a result of the growth in the number and scope of EIAs over the last 30 years. While it is now generally accepted that EIAs tend to increase trade between partner countries, the average partial effects of EIAs on trade flows, reported in papers such as Baier and Bergstrand (2007), Egger et al. (2011) and Anderson and Yotov (2016), are larger than what can be plausibly explained by tariff reductions alone. The combination of the large EIA estimates from the existing literature along with the relatively modest tariff reductions supports the idea that the trade-creating effects from EIAs on bilateral trade are likely to arise from other aspects of the agreements which include the additional provisions and enforceability of the provisions contained in many of these agreements.

One of the potential contributions of this paper is to account for the heterogeneity across agreements in terms of scope and enforceability. As an example of the heterogeneity that exists across agreements, consider the India-Bhutan Free Trade Agreement and the North American Free Trade Agreement. The former does not go beyond commitments in tariff liberalization in goods trade. In contrast, the latter covers commitments in a wide array of topics, including goods and services liberalization, investment liberalization, and environmental and labor standards, to name a few. Additionally, the motives for signing free trade agreements can also differ across country pairs. Rosen (2004) provides evidence that the US-Israel Free Trade Agreement and the US-Jordan Free Trade Agreement are instances where trade policy is used to help pursue foreign policy objectives. Given the heterogeneity in the objectives and design of the agreements, we intend to exploit this heterogeneity and classify agreements in terms of the types of provisions as well as the depth and enforceability of the agreements.

A second potential contribution of the paper is to quantify how this heterogeneity impacts bilateral trade flows using the gravity model of international trade. We use machine learning techniques to identify agreements that are similar in terms of depth of provisions and enforceability and we use these classifications in the gravity model. We find that deeper agreements are typically associated with more trade.

Until recently, the most common approach employed to quantify the impact of EIAs in the empirical gravity literature was to treat the existence of an EIA between trading partners as an indicator variable (cf. Baier and Bergstrand 2007; Anderson et al. 2011; Anderson and Yotov 2016). However, this methodology only measured the average treatment effect and did not take into account the fact that EIAs differ extensively in the scope and the level of integration commitments between the parties.

Baier et al. (2014) provide evidence of the differential effects of different types of trade agreements. They categorize a large number of trade agreements based on their level of economic integration, namely, nonreciprocal preferential trade agreements, reciprocal preferential trade agreements, free trade agreements, customs unions, common markets, and economic unions based on the traditional definition by Frankel et al. (1995, 1997). They find that economic unions and common markets are associated with higher levels of bilateral trade compared with non-reciprocal and reciprocal trade agreements. In short, Baier et al. (2014) were the first to show that more comprehensive trade agreements result in more trade.

Similar to the approach in Baier et al. (2014), Baier et al. (2018), employing a Melitz-style model, provide evidence that heterogeneity exists across the agreements and that the heterogeneity can be associated with lower fixed trade costs versus marginal trade costs. Baier et al. (2019) use a two-stage approach to account for the heterogeneity in trade agreements. In the first stage, they estimate a gravity equation to measure the trade impact by agreement. They then use these estimates to identify economic, geographical, and political factors associated with the agreements.

An alternative approach to the empirical gravity models involves what Kohl (2014) calls a "specialist" approach, in which researchers examine the effect of individual EIAs on the members' trade volumes. In these studies, researchers restrict themselves to a small number of trade agreements in a geographical region with a shorter time horizon. As emphasized in the specialist approach, the commitments in modern trade agreements may go far beyond tariff barriers or factor market integration and incorporate numerous policy areas that may affect the overall bilateral trade costs among member countries. However, these generalizations of these commitments to other trade agreements may be difficult to quantify, and the ability to assess the impact of these commitments on trade may be limited. One of the earliest studies that is related to this line of work was undertaken by Horn et al. (2010), where they examined the content of 14 European Commission and 14 US trade agreements by going through the 28 agreements in their entirety and identifying up to 52 policy areas included in the trade agreements signed by the European Union and the United States. Their work suggests that examining the differential effects of trade agreements requires a detailed analysis of this extensive set of policy areas.

More recently, researchers have combined the empirical gravity approach with the specialist approach. Orefice and Rocha (2014) (referred to as OR hereafter) examine 66 trade agreements from 1980–2007 and use several indices to quantify how the existence and enforceability of provisions influence trade and production networks using the gravity equation. In their work, the provisions are classified as WTO + provisions and include the provisions that are under the current WTO mandate and are already subject to some form of commitment in WTO agreements. The second group of policy areas is WTO-X provisions, which include obligations outside the current WTO mandate. Overall, their study includes 52 policy areas. Kohl et al. (2016) (referred to as KBG hereafter) examine a broader set of trade agreements where they measure the coverage and restrictiveness of policy provisions in trade agreements. They read through 296 trade agreements and quantified them based on the 26 policy areas they cover. In addition, they quantify the legal enforceability of those provisions. For each provision, a score of 0 is assigned if the document does not include the provision, 1 if it does include the provision, and 2 if it includes the provision and the provision is legally enforceable. They then add up those scores, obtain a composite score for each trade agreement, and use this measure of depth in the gravity framework.

Although highly informative and novel in their approach, the OR and KBG studies suffer from a few specification issues. Our paper, using a machine learning approach, mitigates those problems.

First, in OR, they only count the number of legally enforceable provisions. For each provision in each EIA in KBG, they assign scores of 0 if the provision is not included in the agreement, a score of 1 if the provision is present but not enforceable, and a score of 2 if the provision is enforceable. In the KBG study, a score of 2 is assigned as long as a policy provision is covered and if it contains a binding word such as *shall* or *must*, regardless of its specificity and comprehensiveness. In this paper, we use techniques grounded in the distribution of words and phrases in trade agreements to examine the comprehensiveness of coverage areas and their legal enforceability.

Second, with the KBG approach, many provisions must be within the border between 0, 1, and 2. Similarly, in OR where they assigned a binary variable to distinguish between enforceable and not enforceable provisions, there is likely to be some judgment employed. Using machine learning techniques, we will be able to limit the number of judgment calls. In addition, we will be able to capture nuances that the KBG and OR studies might have missed.

Third, because the number of provisions and the correlations of included provisions across EIAs make it difficult to include a set of binary variables for the inclusion of or enforceability of provisions, some sort of data reduction technique needs to be employed to quantify the impact of the provisions. In KBG, they add up the scores across all provisions and obtain a composite score that measures the "depth" of each agreement. Adding up scores across all provisions may not be prudent because not all provisions are trade promoting; some provisions may restrict trade as well, and some provisions are likely bundled together. The KBG approach assumes that each provision is equally important for the pair of countries. However, some provisions may not be necessary for some agreements, and other agreements' provisions may be present, but they may be less complex. For example, North–South trade agreements are more likely to observe labor and environmental provisions that are more detailed than what we might observe in a North–North agreement.[1] OR employ several different approaches to aggregate the data. One approach is to add up the scores and compute a measure of depth similar to KBG. In our approach, we employ a simple form of machine learning that identifies the *clusters* of provisions that are typically grouped together. The appeal of the clustering approach is that it identifies the data patterns commonly associated with different types of agreements. This approach is similar to an alternative approach taken by OR; they use principal component analysis to aggregate the provisions.[2] Principal component analysis forms groups that are linear combinations of the provisions where each group

[1] We recognize that this implies that the provisions' determination and complexity are endogenously determined. However, the theoretical and empirical determination of what provisions are included in a trade agreement is beyond the scope of this paper.

[2] See Ding and He (2004) for the similarities between K-means clustering and principal component analysis.

is orthogonal. Because the groups are linear combinations of the provisions, it can be difficult to interpet the results or use the results for ex-ante prediction. In our approach, we can observe the likelihood of each provision in each cluster. We think our results are easier to interpret and our approach makes it easier to identify what provisions are associated with higher bilateral trade flows.

Finally, the KBG study examines only the coverage and the enforceability of 26 trade policy areas, whereas our study examines 36 policy areas. OR include more provisions but cover a shorter time horizon, and their focus is on production networks (parts and components).

In addition to the papers mentioned above, this paper shares similarities with two recent studies. Mattoo et al. (2022) cover the same period (1970–2014) and nearly the same number of agreements, 280 in our study to 279 in theirs. The main difference is that their approach follows KBG in measuring the enforceability and depth of the agreements. More recently, Breinlich et al. (2021) use a non-linear lasso (and iceberg lasso) as a variable selection approach to dealing with correlation and dimensionality of provisions. While their paper identifies the provisions that are associated with higher trade flows, it is not possible to infer that these are the provisions that matter the most for trade. Similarly, our approach to clustering the data does not tell how each provision impacts bilateral trade flows. However, our empirical results inform us of which provisions, when bundled together, are associated with higher bilateral trade flows. In this sense, we view the two studies as complementary. As in Breinlich et al. (2021), we find agreements that include technical barriers to trade, antidumping, trade facilitation and competition policies to be associated with higher trade flows.

The rest of the paper proceeds as follows. First, we classify trade agreements into distinct clusters using *k*-means clustering, an unsupervised learning method. Unsupervised learning is a machine learning technique that identifies patterns in datasets without users having to provide any labels. In contrast, supervised learning methods train on labeled data to figure out patterns in new data.

Second, we use multilabel classification, a supervised learning method, to examine the nature of each cluster for the coverage and comprehensiveness of specific trade policies of interest. It turns out that the groupings of trade agreements obtained from clustering in the first stage carry economic interpretation. The clustering exercise can separate shallow agreements from deep agreements quite well.

Third, we run the gravity model of trade regression and find evidence that trade agreements that cover a wide range of trade policy areas with high legal enforceability lead to the most substantial impact on trade flows. Our approach also allows us to identify the provisions that have the most substantial impact on trade creation. These provisions are antidumping, capital mobility, competition laws, customs harmonization, dispute settlement framework, e-commerce, environment, export and import restrictions, freedom of transit, investment, labor, public procurement, sanitary and phytosanitary measures, services, subsidies and countervailing measures, technical barriers to trade, telecommunications liberalization, and transparency. These results have far-reaching implications in shaping modern trade policy institutions.

The paper is structured as follows. Section 2 explains the data used in the analysis. Sections 3 and 4 present discussions on clustering and classification, respectively.

Section 5 discusses the gravity model of trade flows as applied in the context of our study. Section 6 provides the main empirical results and findings, section 7 provides robustness analysis, and section 8 concludes.

2 Database of Trade Agreements

We use a total of 280 trade agreement texts in our analysis. The trade agreement documents and data on relevant bilateral country pairs come from Baier and Bergstrand (2017).[3] The economic integration agreement (EIA) database is a panel dataset that includes bilateral agreements between pairs of countries annually from 1950 to 2012.[4] We combine these data with bilateral trade data from Comtrade so that we have 159 countries in our study, which implies that we have approximately 25,000 country pairs annually from 1970 to 2014. For each bilateral pair, the dataset maps the relevant EIA (along with the pdf document of the actual treaty) governing the bilateral relationship and when the EIA status of the bilateral pair changes.[5] We remove annexes, protocols, and schedules and focus on the text's main body to ensure that our clustering results are not driven by the mere presence of schedules, protocols, and annexes.

3 Clustering

The application of machine learning methods requires converting trade agreement texts to numerical vectors. We remove punctuation, symbols, numbers, and white spaces and segment trade agreement texts into single words and two-word phrases, also known as bigrams in the clustering literature. We then count the frequency of single word and two-word phrases within a trade agreement and normalize the frequency by a vector of normalized frequencies of single-word and two-word phrases.[6]

We then count the frequency of single word and two-word phrases within a trade agreement and normalize the frequency by the document's size. Each trade agreement is uniquely represented by a vector of normalized frequencies of single words and two-word phrases.[7]

The goal of cluster analysis is to find natural groupings of objects based on a number of object features. The first stage of clustering involves determining the "appropriate" number of clusters, and the second stage involves grouping the trade agreements into the "appropriate" number of clusters. We use one of the most

[3] These data are similar to the database used in the KBG study. The main difference is that these data cover more periods. For each agreement, there are hyperlinks to associated pdf documents that contain the terms of the agreement or modifications to the agreement.

[4] Appendix A lists all the countries used in the empirical analysis.

[5] Appendix B lists the bilateral agreements for each pair and the years in force.

[6] We also conduct the exact analysis done in the main body of this paper in the robustness section by removing extremely rare and common words and two-word phrases, and the main results still hold.

[7] Appendix C discusses the methods in detail.

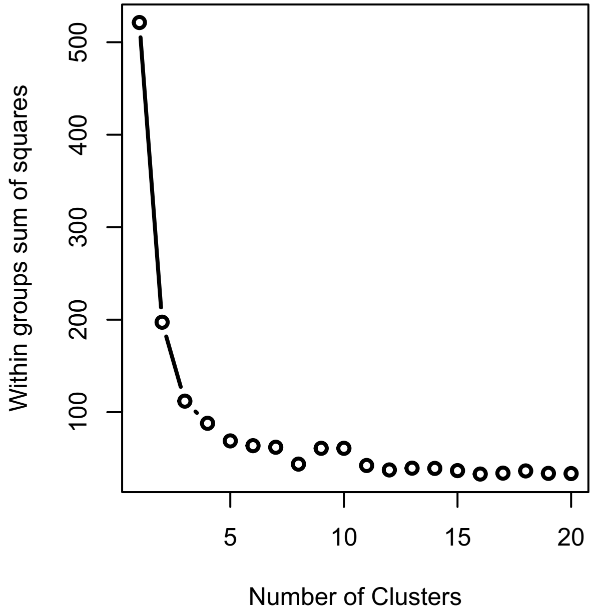

Fig. 1 Within-Groups Sums of Squares

widely used clustering techniques called the k-means algorithm due to Hartigan and Wong (1979) and apply standard procedures in choosing the appropriate number of clusters by minimizing the sum of squared errors between the empirical mean of a cluster and the objects assigned to that cluster over all clusters.

We will now formally introduce the k-means algorithm. Let $X=\{x_i\}$, $x_i \in \mathrm{R}^{\mathrm{D}}$ be the set of trade agreements to be clustered into a set of K clusters, $C=\{c_k: k=1,..,K\}$, where D is the number of features per trade agreement. Assume a priori that there exist K clusters with cluster centers $\mu_1, \mu_2, \ldots, \mu_k \in \mathbb{R}^D$.[8] K-means clustering solves

$$argmax_C \sum_{i=1}^{K} \sum_{x \in C_i} \|x_i - \mu_i\|^2. \quad (1)$$

The implementation of this optimization takes place in the following steps:

1. Choose an initial number of clusters, k, based on domain knowledge.
2. Initialize cluster centers $\mu_1, \mu_2, \ldots, \mu_k$ arbitrarily.
3. Given the fixed cluster centers, choose the optimal group assignment for each data point (trade agreements) x_i on the basis of the closest cluster center.
4. Update $\mu_1,\mu_2,\ldots,\mu_k$ on the basis of group assignments of x_i.
5. Repeat steps 3 and 4 until convergence—that is, until the cluster's centroids do not move.

We repeat the preceding steps for different values of K and choose an appropriate number of clusters, $k*$.

[8] Although we assume that there exist K centers, we will eventually update this on the basis of the value of our loss function and the application in hand.

Table 1 Correlation between the Two Sets of Clusters

	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Cluster 1	**0.42**	−0.21	−0.16	−0.22
Cluster 2	**0.65**	−0.33	−0.25	−0.35
Cluster 3	−0.51	**1.00**	−0.13	−0.18
Cluster 4	−0.39	−0.13	**1.00**	−0.14
Cluster 5	−0.54	−0.18	−0.14	**1.00**

This table shows the correlation matrix between the two sets of clusters. As we move from five clusters to four clusters, cluster 1 and cluster 2 merge into one cluster, whereas the rest of the clusters do not change, as indicated by a correlation of 1.00

(A) **Optimal Number of Clusters**

We use one of the most commonly used techniques in the literature to determine the appropriate number of clusters. The technique is known as the "elbow method," where within-groups sums of squares are plotted against the number of clusters. If the plot resembles an arm, then the "elbow" on the arm is the appropriate number of clusters. Figure 1 plots the within-groups sums of squares against the number of clusters. As suggested by the elbow method, the appropriate number of clusters is anywhere between four and five. The within-group sum of squares does not fall much after the fifth cluster, implying that the upper bound on the number of clusters is five. Throughout the rest of the paper, we will present the results when trade agreements are split into four and five clusters.

(B) **Five Clusters vs. Four Clusters**

As we move from five clusters to four clusters, it is natural for a few trade agreements to change their cluster memberships. However, it would be a cause for concern if many agreements switched their cluster memberships. To investigate the movement of country pairs between clusters, we report the correlation between the two sets of clusters. It turns out that the clusters appear to be relatively stable. Table 1 shows the correlation matrix between the two sets of clusters. As we move from five clusters to four clusters, most of the trade agreements stay in their "natural" groups, as indicated by the perfect correlation (correlation = 1.00) in three of the cluster groups. The only time we see movements in cluster assignments is when the pairs in clusters 1 and 2 (in the case with five clusters) merge into a single cluster, and the rest of the clusters' membership does not change.[9]

(C) **Stability of Clustering Results**

We ensure the stability of our clustering results by running 1,000 iterations of *k*-means clustering in the cases of both four and five clusters. We then compute the correlation between the 1,000 sets of clustering results against the base clustering result used in this paper.

In the case of four clusters, the mean correlation is 0.9796, and the standard deviation is 0.0656. This outcome indicates that clustering results are highly

[9] We could also use seven clusters as indicated by the elbow method; however, as we will discuss, the four and five clusters provide a cleaner economic interpretation.

stable, and trade agreements do not change cluster membership erratically. The correlation is also relatively high in the case of five clusters. The mean correlation is 0.9294, and the standard deviation is 0.1513. Evidently, the lower correlation and higher standard deviation occur with five clusters as some of the pairs switch between clusters 1 and 2. Therefore, this result makes us confident that the *k*-means clustering is a promising way to find natural grouping in the trade agreements.

4 The Characteristics of the Identified Clusters

Thus far, we have gathered trade agreements into their natural groupings. However, we have not yet presented any insights into the actual content of the trade agreements in each cluster. We now turn to a discussion of the supervised learning method, which enables us to analyze the content of trade agreements in each cluster.

(A) **Supervised Learning: Classification**

Supervised learning involves inferring the underlying function from labeled training data. The training data comprise a set of training examples and a label associated with each training example. On the basis of the training examples and the user-specified labels for these examples, the supervised learning method infers the label for test data. More formally, given training examples of the form $(x_1,y_1),\ldots(x_N,y_N)$, where x_i is a vector of features and y_i is an assigned label, the goal of a learning algorithm is to infer a function g: $X \rightarrow Y$ and thus to predict the output label for an unseen test sample.

In the context of our study, we train our model with instances of 36 trade policy provisions and allow the machine learning algorithm to "crawl" through each paragraph of the trade agreement to estimate the likelihood of the paragraph being about one or more policy areas. The 36 policy areas we identify are agriculture, anticorruption, antidumping, capital mobility, competition, consumer protection, customs administration, dispute settlement, e-commerce, education and training cooperation, energy, environment, financial cooperation, freedom of transit, export and import restrictions, industrial cooperation, institutional arrangements, intellectual property rights, investment, investor-state dispute settlement, labor, money laundering and illicit drugs, political dialogue, public procurement, safeguard procedures, sanitary and phytosanitary measures, science and technology, services, small and medium-sized enterprises, state aid, state trading enterprises, subsidies and countervailing measures, technical barriers to trade, telecommunications, transparency, and transportation infrastructure.

We train the model with examples of comprehensive and legally binding provisions for each policy area. For policy areas already covered in the World Trade Organization (WTO) agreements, we seek commitments above and beyond the WTO agreements. For policy areas that are not already a part of the WTO agreements, we compare the provisions in all trade agreements and choose the most comprehensive ones. By legally binding, we mean that the provision is very specific; the provision contains at least one restrictive word, such as *shall* or *must*. The provision specifies the course of action in case either party deviates from the commitment listed in the provision. This course of action may be different from a generic dispute settlement present in the trade agreements. The following provision is a training example for the *Investment* provision:

National Treatment

1. *Each Country shall accord to investors of the other Country and to their investments treatment no less favorable than that it accords in like circumstances to its own investors and to their investments with respect to the establishment, acquisition, expansion, management, operation, maintenance, use, possession, liquidation, sale, or other disposition of investments (hereinafter referred to in this Chapter as investment activities). Each Country shall accord to investors of the other Country and to their investments treatment no less favorable than that it accords in like circumstances to investors of a third State and to their investments, with respect to investment activities.*

Article 9.6: Performance Requirements

Neither Party may impose or enforce any of the following requirements, enforce any commitment or undertaking, in connection with the establishment, acquisition, expansion, management, conduct, operation, or sale or other disposition of an investment of an investor of a Party or of a non-Party in its territory to:

(a) *export a given level or percentage of goods or services;*
(b) *achieve a given level or percentage of domestic content;*
(c) *purchase, use, or accord a preference to goods produced in its territory, or to purchase goods from persons in its territory;*
(d) *relate in any way the volume or value of imports to the volume or value of exports or to the amount of foreign exchange inflows associated with such investment;*
(e) *restrict sales of goods or services in its territory that such investment produces or provides by relating such sales in any way to the volume or value of its exports or foreign exchange earnings;*
(f) *transfer a particular technology, production process, or other proprietary knowledge to a person in its territory;*
(g) *supply exclusively from the territory of the Party the goods that it produces or the services that it provides to a specific regional market or to the world market.*

The preceding paragraph is highly comprehensive about investment commitments between the parties, and the language of the text is also highly enforceable. We feed training examples such as the one above for each of the 36 provisions and perform a multilabel classification on each paragraph of trade agreements to estimate the paragraph's likelihood of being associated with one or more than one category. Essentially, we find a similarity measure between an untrained paragraph and the trained examples. Because a paragraph can be related to more than just one policy domain, we also allow for multilabel classification with a technique called cross-training (Boutell et al. 2004). The idea is to use paragraphs with multiple labels more than once during training. This implies that each training example can be a positive instance for more than one category during training. We then perform multilabel classification as 36 individual binary classification problems using a one-vs.-rest strategy. This method has proved effective in classifying multilabel images in the classification literature. Therefore, for each unit of analysis (i.e., a paragraph), we estimate the probability of the unit being relevant in one or more of the 36 categories. As is common in the literature, we put a threshold probability of 0.5 for the document to both be pertinent about the category and to have a minimum level of legal enforceability in its language (McLaughlin and Sherouse 2016).

We experimented with the two popular classification algorithms: *k*-nearest neighbors and random forest classifier. We evaluated both models' performance using macro F1-scores averaged across all classes, the details of which will be presented in the next subsection. This process reveals that the *k*-nearest neighbors classifier performs superior to the random forest classifier. We use *k*-nearest neighbors with five neighbors and uniform weights to classify our trade agreement paragraphs. With this, a brief discussion of the *k*-nearest neighbors classifier is thus warranted.

(B) ***K*-Nearest Neighbors Classifier**

The k-nearest neighbors classifier is one of the most popular nonparametric classification algorithms in many machine learning applications. Given training data $D=(x_1,y_1),\ldots,(x_N,y_N)$ and a positive integer k where x_i is a vector of features and y_i represents self-assigned labels for that set of features, the class prediction for a new test point x_0 involves identifying the K observations in the training data that are closest to x_0. Therefore, the conditional probability for class j is given as

$$Pr\ (Y=j|X=x)=\frac{1}{K}\sum_{\Omega} I(y_i=j) \tag{2}$$

where Ω represents the set of K observations closest to the x_0 (Friedman et al. 2001).

(C) **Model Evaluation**

We evaluate our models and choose the appropriate parameters by performing classification on fivefold cross-validated data. Cross-validation is a model validation technique to assess the generalizability of the classification results. The idea is that the training examples are further separated into training and test set, and the classification model is estimated only on the training set. The model is then used to predict the class of the test set.

The metric used to evaluate the classification model is the macro F1-scores averaged across all classes. F1-scores are the harmonic average of precision and recall. In a binary classification setting, precision is the percentage of selected items that are correct. So, precision is given by

Table 2 Macro F1-Scores by Model: Classification

Model	Average Macro F1-Scores
K-nearest neighbors, weight = uniform and k = 7	0.8366 (0.0696)
K-nearest neighbors, weight = distance and k = 7	0.8160 (0.0704)
K-nearest neighbors, weight = uniform and k = 6	0.8243 (0.0632)
K-nearest neighbors, weight = distance and k = 6	0.8214 (0.0635)
K-nearest neighbors, weight = uniform and k = 5	0.8373 (0.0649)
K-nearest neighbors, weight = distance and k = 5	0.8123 (0.0649)
Random forest classifier, number of trees = 5	0.5712 (0.0704)
Random forest classifier, number of trees = 10	0.5388 (0.0635)
Random forest classifier, number of trees = 15	0.5991 (0.0628)

F1-scores are averaged across all classes using fivefold cross-validation. For k-nearest neighbors model, k represents the number of neighbors used. The weight parameter of distance indicates that within a class, closer neighbors will have a greater influence than neighbors that are farther away

$$\frac{True\ Positives}{True\ Positives + False\ Positives.} \tag{3}$$

Similarly, recall is defined as the percentage of correct items that are selected. So, recall is given by

$$\frac{True\ Positives}{True\ Positives + False\ Negatives.} \tag{4}$$

Then F1-score is given by

$$F1\ score = \frac{2 \times Precision \times Recall}{Precision + Recall\ \ .} \tag{5}$$

This measure strikes a balance between precision and recall. For our classification, we average F1-scores across all classes. Table 2 presents average F1-scores across all classes from fivefold cross-validation for k-nearest neighbors and the random forest classifier with various parameter values. It can be seen that the k-nearest neighbors classifier performs better than the random forest classifier for every parameter value tested. Within the *k*-nearest neighbors' classifier, the model with five neighbors and uniform weight performs the best classification, which we use to classify our test data.

(D) **Provision Scores**

Table 3 reports the relative frequencies of 36 provisions across the trade agreements in our sample. We place a threshold score of 0.5 for a provision to be counted in a trade agreement in this table. The most common provision is the liberalization of export and import restrictions with more than 95 percent of the trade agreements containing at least a minimum level of content as well as legal enforceability. This result seems obvious because the purpose of most trade agreements is to remove tariffs and other import restrictions. The next four common provisions are on safeguard procedures, competition, intellectual property rights, and agriculture, respectively, with each present in more than half of the agreements. Similarly, the least common provision is consumer protection, with only about 5 percent of the trade agreements containing this provision. The next five least common provisions are anticorruption, transport infrastructure, energy, and money laundering and illicit drugs, respectively. Each is present in eight percent or fewer of the trade agreements in our sample.

We also perform correlation on all of our provision scores simultaneously to understand what provisions tend to co-occur in trade agreements. Figure 2 represents the correlation matrix heatmap. Dark blue squares represent a very high positive correlation, dark red squares represent a very high negative correlation, and white squares represent zero correlation. We reorganize the figure so that

Table 3 Most Common to Least Common Provisions

Provision % of FTAs		Provision % of FTAs	
Export/import restrictions	95	Services	26
Safeguard procedures	71	Investment	23
Competition	57	Investor-state dispute settlement	19
Intellectual property rights	57	Political dialogue	18
Agriculture	53	Freedom of transit 17	17
Dispute settlement	49	Science and technology cooperation	17
State trading enterprises	47	Telecommunications	16
Institutional Arrangements	44	Environment	14
Technical barriers to trade	42	E-commerce	13
Public procurement	41	Industrial Cooperation	12
Customs administration	38	Education and training	11
Capital mobility	38	Financial cooperation	10
State aid	36	Small and medium-sized enterprises	9
Transparency	35	Money laundering/illicit drugs	8
Sanitary/phytosanitary measures	33	Energy	7
Antidumping	32	Transportation infrastructure	7
Labor	29	Anticorruption	6
Subsidies/countervailing measures	29	Consumer protection	5

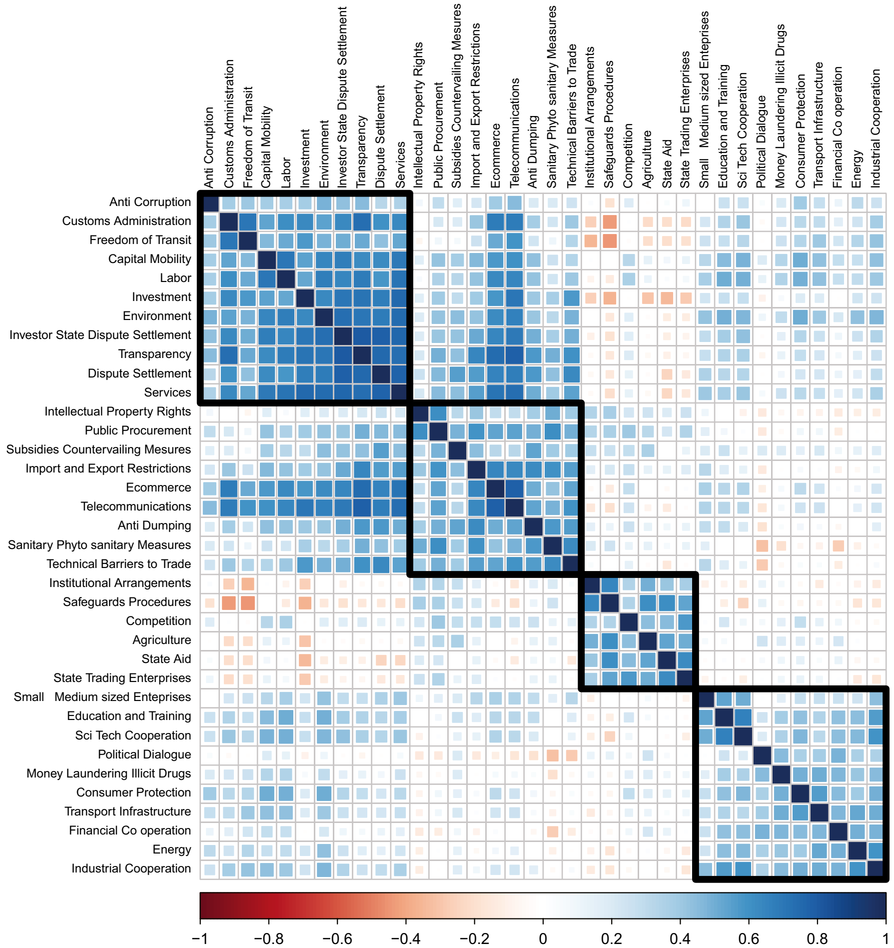

Fig. 2 Heatmap: Correlation Across Provisions

the provisions that are more likely to co-occur are grouped together in the boxes outlined in black. For instance, provisions on anticorruption, customs administration, freedom of transit, capital mobility, labor, investment, environment, investor-state dispute settlement, transparency, dispute settlement (state-state), and services are more likely to be present together in trade agreements, as seen in the topmost left square. It turns out that according to our gravity analysis, which is the topic of the next section, most of these provisions, when bundled together, are highly trade promoting. Similarly, provisions on intellectual property rights, public procurement, subsidies and countervailing measures, export and import restrictions, e-commerce, antidumping, sanitary and phytosanitary measures, and technical barriers to trade do co-occur in trade agreements, as do provisions on institutional arrangements, safeguard procedures, competition, agriculture, state aid, and state trading enterprises.

Furthermore, there is strong support for provisions on political dialogue, consumer protection, measures against money laundering and illicit drugs, financial cooperation, energy, and industrial cooperation to co-occur together, as seen in the bottom right square. These provisions are commonly present in trade agreements written by the European Union with African economies and some eastern European countries before they joined the European Union. On an individual provision level, some relationships are worth mentioning. For example, provisions on transparency and harmonization of customs administration tend to generally feature together. The same goes for services and investment provisions. However, state aid and investment provisions tend not to feature in the same trade agreement.

Recall that the *k*-means clustering algorithm groups the trade agreements into different clusters without any intent of subscribing economic meaning to the clusters. However, given the provision scores for each cluster, it is easy to label the clusters for the depth of the agreements. Tables 4 and 5 show that these clusters can be organized by the depth of the trade agreement going from the least comprehensive (labeled “Shallowest”) to the most comprehensive agreements (labeled “Deepest”). Thus, the *k*-means clustering groups by the depth of the agreement, whereby a deeper agreement implies that the cluster is more likely to have higher provision scores than a shallow agreement. One might be inclined to think these clusters map directly into the depth measures used in Hofmann et al. (2019). We compare our clusters to the depth measure in Hofmann et al. (2019) and find that the correlation is 0.7804. The main difference in the coding is that our approach goes beyond the 0, 1, 2 coding and considers multiple components of the provisions, such as specificity and comprehensiveness of the provision, references to international agreement relevant to the provision, the presence, or lack thereof, of institutional arrangements as a forum for addressing and handling disagreements on that provision, as well as the indication of a time frame for various instances. As in Hofmann et al. (2019), we find that the agreements' depth has increased over time as presented in Fig. 3. The bars represent the number of FTAs signed in each year. Each bar is then split into five categories based on our cluster classification of “Shallowest” (red) to “Deepest” (dark green). The proportion of more comprehensive agreements relative to all agreements signed in that year increases over time, suggesting that agreements signed in recent years tend to include more policy areas with higher legal enforceability than older agreements. Especially, agreements signed after the early to mid-2000s tend to be significantly more comprehensive than older agreements.

Similarly, we compare our depth measure to coding in Baier et al. (2014). Recall that in their study, the authors classify the agreements by FTAs, customs unions, common markets, and economic unions. We code these agreements from the shallowest (FTA) to the deepest economic union and compare these to our measure of depth. Tables 6 and 7 show the correlation between these definitions and the clusters identified in this paper. Although the clusters used in this paper,

Table 4 Provision Scores for Five Clusters

Provision	Shallowest	Shallow	Moderate	Deep	Deepest
Agriculture	0.25	0.70	0.66	0.67	0.49
Anticorruption	0.00	0.00	0.02	0.37	0.15
Antidumping	0.08	0.27	0.52	0.37	0.70
Capital mobility	0.09	0.25	0.55	0.87	0.89
Competition	0.19	0.51	0.49	0.57	0.70
Consumer protection	0.00	0.016	0.013	0.28	0.16
Customs administration	0.26	0.14	0.69	0.85	0.90
Dispute settlement	0.27	0.36	0.78	0.82	0.92
E-commerce	0.00	0.02	0.18	0.56	0.44
Education & training cooperation	0.02	0.04	0.21	0.16	0.30
Energy	0.03	0.06	0.13	0.17	0.17
Environment	0.05	0.09	0.22	0.65	0.51
Export & import restrictions	0.69	0.85	0.91	0.92	0.93
Financial cooperation	0.02	0.06	0.25	0.19	0.19
Freedom of transit	0.17	0.06	0.29	0.59	0.48
Industrial cooperation	0.09	0.10	0.24	0.24	0.30
Institutional arrangements	0.34	0.90	0.75	0.64	0.73
Intellectual property rights	0.28	0.77	0.57	0.70	0.70
Investment	0.10	0.12	0.34	0.74	0.70
Investor-state dispute settlement	0.04	0.07	0.29	0.78	0.62
Labor	0.03	0.06	0.46	0.84	0.87
Money laundering/illicit drugs	0.00	0.05	0.19	0.11	0.15
Political dialogue	0.14	0.22	0.29	0.25	0.19
Public procurement	0.05	0.45	0.47	0.77	0.64
Safeguard procedures	0.21	0.90	0.71	0.66	0.65
Sanitary & phytosanitary measures	0.13	0.38	0.45	0.68	0.64
Science & technology cooperation	0.10	0.07	0.27	0.40	0.43
Services	0.07	0.10	0.44	0.74	0.80
Small & medium-sized enterprises	0.03	0.05	0.19	0.15	0.20
State aid	0.16	0.51	0.39	0.48	0.40
State trading enterprises	0.19	0.45	0.68	0.63	0.52
Subsidies & countervailing measures	0.08	0.38	0.48	0.44	0.45
Technical barriers to trade	0.23	0.34	0.64	0.71	0.78
Telecommunications	0.01	0.00	0.18	0.70	0.56
Transparency	0.00	0.06	0.60	0.91	0.88
Transport infrastructure	0.01	0.04	0.15	0.18	0.19

Table reports the average provision scores across trade agreements within each cluster for five clusters. Low provision scores across all clusters implies that the provision is only prevalent in a few trade agreements

Table 5 Provision Scores for Four Clusters

Provision	Shallowest	Moderate	Deep	Deepest
Agriculture	0.54	0.66	0.67	0.49
Anticorruption	0.00	0.02	0.37	0.15
Antidumping	0.21	0.52	0.37	0.70
Capital mobility	0.19	0.55	0.87	0.89
Competition	0.40	0.49	0.57	0.70
Consumer protection	0.01	0.13	0.28	0.16
Customs administration	0.18	0.69	0.85	0.90
Dispute settlement	0.33	0.78	0.82	0.92
E-commerce	0.01	0.18	0.56	0.44
Education & training cooperation	0.03	0.21	0.16	0.30
Energy	0.05	0.13	0.17	0.17
Environment	0.07	0.22	0.65	0.51
Export & import restrictions	0.79	0.91	0.92	0.93
Financial cooperation	0.05	0.25	0.19	0.19
Freedom of transit	0.10	0.29	0.59	0.48
Industrial cooperation	0.10	0.24	0.24	0.30
Institutional arrangements	0.71	0.75	0.64	0.73
Intellectual property rights	0.60	0.57	0.70	0.70
Investment	0.11	0.34	0.74	0.71
Investor-state dispute settlement	0.06	0.29	0.78	0.62
Labor	0.05	0.46	0.84	0.87
Money laundering/illicit drugs	0.04	0.19	0.11	0.15
Political dialogue	0.20	0.29	0.25	0.19
Public procurement	0.31	0.47	0.77	0.64
Safeguard procedures	0.66	0.71	0.66	0.64
Sanitary & phytosanitary measures	0.30	0.45	0.68	0.64
Science & technology cooperation	0.08	0.27	0.40	0.43
Services	0.09	0.44	0.74	0.80
Small & medium-sized enterprises	0.04	0.19	0.15	0.20
State aid	0.39	0.39	0.48	0.40
State trading enterprises	0.51	0.45	0.63	0.52
Subsidies & countervailing measures	0.28	0.48	0.44	0.45
Technical barriers to trade	0.30	0.64	0.71	0.78
Telecommunications	0.01	0.18	0.70	0.56
Transparency	0.04	0.60	0.91	0.88
Transport infrastructure	0.03	0.15	0.18	0.19

Table reports the average provision scores across trade agreements within each cluster for four clusters. Low provision scores across all clusters implies that the provision is only prevalent in a few trade agreements

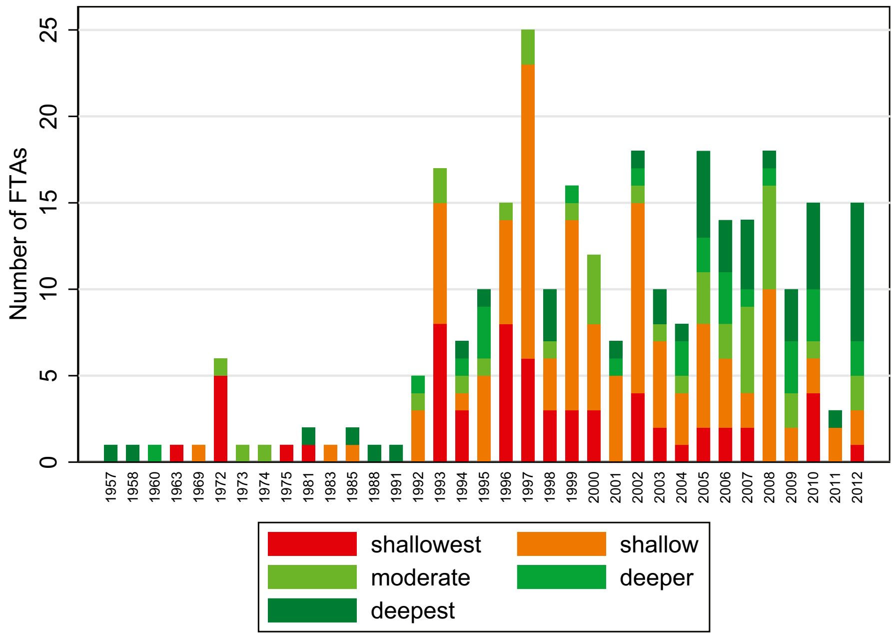

Fig. 3 Depth of Trade Agreements over Time. Note: The greener the bar in that year, the deeper the agreements signed during that year

organized by depth, show some degree of correlation with the types of agreements in Baier et al. (2014), the correlation is far from perfect.

5 Gravity Analysis with Cluster

In section III, we cluster trade agreements in their natural groupings. In the subsequent section, we describe the actual contents of trade agreements belonging to each cluster. Here, we combine the cluster membership information obtained from the

Table 6 Correlation Clusters and Types of Economic Integration Agreements

	FTA	CU	CM	EU
Shallowest	0.399	0.055	−0.007	0.078
Shallow	0.617	0.074	−0.011	−0.008
Moderate	0.274	0.332	0.248	0.110
Deep	0.126	−0.009	0.575	0.432
Deepest	0.239	0.431	0.004	−0.006

This table shows the correlation matrix between the clusters and the EIAs used by Baier et al. (2014)

FTA free trade agreement, *CU* customs union, *CM* common market, *EU* Economic Union

Table 7 Correlation Clusters and Types of Economic Integration Agreements

	FTA	CU	CM	EU
Shallowest	0.737	0.093	−0.013	0.040
Moderate	0.274	0.332	0.248	0.110
Deep	0.126	−0.009	0.575	0.432
Deepest	0.239	0.431	0.004	−0.006

This table shows the correlation matrix between the clusters and the EIAs used by Baier et al. (2014)

FTA free trade agreement, *CU* customs union, *CM* common market, *EU* economic union

first stage with gravity analysis of trade flows to obtain more insights into the heterogeneous impacts of FTAs on trade flows. The gravity regression results—along with the clustering and classification procedures performed in the previous sections—will enable us to answer policy-relevant questions, such as what provisions or set of provisions matters the most for trade flows and under what conditions.

The gravity model—often referred to as the workhorse model in international trade—is widely used to study the effects of various determinants of country pairs' goods and factor flows. The gravity model uses the metaphor of Newton's law of gravitation and predicts that the trade flow between two countries is directly proportional to the product of their economic mass, typically measured by GDP, and inversely proportional to the distance between the two countries. Tinbergen (1962) uses the gravity equation to evaluate the effects of FTA dummy variables on bilateral trade flows among European countries. Despite its empirical success, the initial applications were atheoretical. Anderson (1979) conducted the first study to provide microeconomic foundations to the gravity equation by employing a model where goods were specific to the country of origin and agents had "Armington" preferences over goods from each country. Since then, other studies have shown that "gravity-like" relationships emerge in various settings (Bergstrand 1985; Eaton and Kortum 2002; Melitz 2003; Anderson and van Wincoop 2003).

Following Feenstra (2006) and Baier and Bergstrand (2007), we present the following gravity specification:

$$\ln\left(X_{ij,t}\right) = \delta_{i,t} + \delta_{j,t} + \chi_{ij} + \sum\nolimits_{k=1}^{K} FTA_{ij,t} * cluster_k + \epsilon_{ij,t}, \tag{6}$$

where the $lnX_{ij,t}$ represents the logarithm of trade flows from country i to country j, $\delta_{i,t}$ is the vector of exporter-time fixed effects that captures any exporter-specific factors, $\delta_{j,t}$ is the vector of importer-time fixed effects that captures any importer-specific factors, and the vector χ_{ij} denotes time-invariant country-pair fixed effects.

These importer-time and exporter-time fixed effects control for "multilateral resistance," first introduced by Anderson and van Wincoop (2003) and all other country-time specific variables related to the exporter and importer. As discussed in Baier and Bergstrand (2007), we include country-pair fixed effects to account any possible endogeneity of free trade agreements. $FTA_{ij,t}$ is a dummy variable that takes a value of 1 if country i and country j have an FTA between them and a value of 0 otherwise. $Cluster_k$ is also a dummy variable and only takes a value of 1 if $FTA_{ij,t}$ belongs to cluster k, and K is the total number of clusters.

One of the potential drawbacks of specification (6) is that, in the presence of heteroskedasticity, the coefficient estimates are likely to be biased and inconsistent, as pointed out by Santos Silva and Tenreyro (2006), hereafter SST. They suggest using a Poisson pseudo-maximum likelihood (PPML) estimator. Following SST, we also estimate

$$X_{ij,t} = \exp\left[\delta_{i,t} + \delta_{j,t} + \chi_{ij} + \sum\nolimits_{k=1}^{K} FTA_{ij,t} * Cluster_k\right] + \epsilon_{ij,t} \tag{7}$$

We estimate specifications (6) and (7) for two sets of clustering results: one for five clusters and one for four clusters, where we do not remove overly common and rare words.

6 Results and Discussion

After performing the preceding text mining exercises along with the gravity analysis of trade flows, we have a compelling way to answer the following questions: Do more comprehensive agreements result in more trade creation? What trade policy provision or set of trade policy provisions matters the most for trade flows? To compare our results to previous work, Table 8 presents some baseline results. Columns 1 and 2 provide baseline estimates of the gravity equation when we use the definition of trade agreements in Baier and Bergstrand (2007) and Anderson and Yotov (2016). In column 1, the results from the standard baseline OLS model with an indicator variable for a trade agreement indicate that the presence of a trade agreement increases trade by about 50 percent (=(exp(0.401)-1)*100). For the PPML specification, the trade agreements boost bilateral trade by about 11 percent. If we use KBG data for provisions, we see that the coefficient on depth is 0.406 for OLS; therefore, if a country pair had all of the provisions and they were all enforceable (depth = 1), trade would be expected to be about 50 percent higher. If they had a depth measure of 0.50, which is close to their mean measure of depth, trade would

Table 8 Baseline Gravity Regressions

Cluster	OLS FE FTA	PPML FE	OLS FE	PPML FE
FTA	0.401*** (0.015)	0.108*** (0.015)		
KBG depth			0.406*** (0.020)	0.077*** (0.011)
$R^{-}2$	0.8296	-	0.8297	-
N	611,948	611,948	611,948	611,948

Table reports the coefficients on the gravity regressions with country-pair fixed effects, importer fixed effects, and exporter fixed effects

***1%, **5%, *10%

Table 9 Gravity Regressions for Four and Five Clusters

Cluster	OLS FE 5 clusters	PPML FE 5 clusters	OLS FE 4 clusters	PPML FE 4 clusters
Shallowest	0.275*** (0.035)	0.118*** (0.015)	0.319*** (0.020)	0.096*** (0.011)
Shallow	0.336*** (0.022)	0.083*** (0.013)	-	-
Moderate	0.321*** (0.025)	0.045*** (0.013)	0.322*** (0.025)	0.046*** (0.013)
Deep	0.622*** (0.028)	0.138*** (0.119)	0.624*** (0.028)	0.137*** (0.012)
Deepest	0.630*** (0.031)	0.161*** (0.012)	0.632*** (0.031)	0.158*** (0.012)
$\bar{R}2$	0.8296	-	0.8297	-
N	611,681	611,681	611,681	611,681

Table reports the coefficients on the gravity regressions with country-pair fixed effects, importer fixed effects, and exporter fixed effects

***1%, **5%, *10%

be about 23 percent higher. Using the KBG measure and using PPML, we find that the deepest agreements would increase trade by about eight percent and the increase in trade evaluated at the mean of the depth measure would be about four percent.

Turning to our data with four and five clusters, the first column of Table 9 presents the gravity trade regression results when trade agreements are grouped into five clusters where the model is estimated using OLS. This result indicates that the most comprehensive set of agreements, denoted by *deepest*, has the largest impact on trade flows. The coefficient for this cluster is about 0.630 for OLS. It indicates that the most comprehensive trade agreements can increase two members' trade flows by nearly 88 percent (=(exp(0.630)-1)*100=87.8%). With five four and five clusters, we find the coefficients on the deepest agreements are roughly twice the size of the coefficients on the shallower agreements. The results are not quite as dramatic using PPML. Deeper agreements are associated with more trade but the magnitude of the coefficient is smaller.[10] It is also noteworthy that, unlike the OLS estimates, the coefficient on the agreement does not increase with depth. With five clusters the shallowest agreements are associated with higher bilateral trade than the agreements we label as shallow and moderate. With four clusters, the shallowest agreement is associated with higher bilateral trade than moderate agreements. For the PPML in both four and five clusters, the deep and deepest agreements are associated with higher bilateral trade. One potential reason for this may be because of the potential endogeneity of the depth of the agreements the countries choose. While it is difficult to make direct comparisons to

[10] Although we only report results with only positive trade flows in this paper, we have also estimated PPML with the inclusion of zeros. These results are available on request.

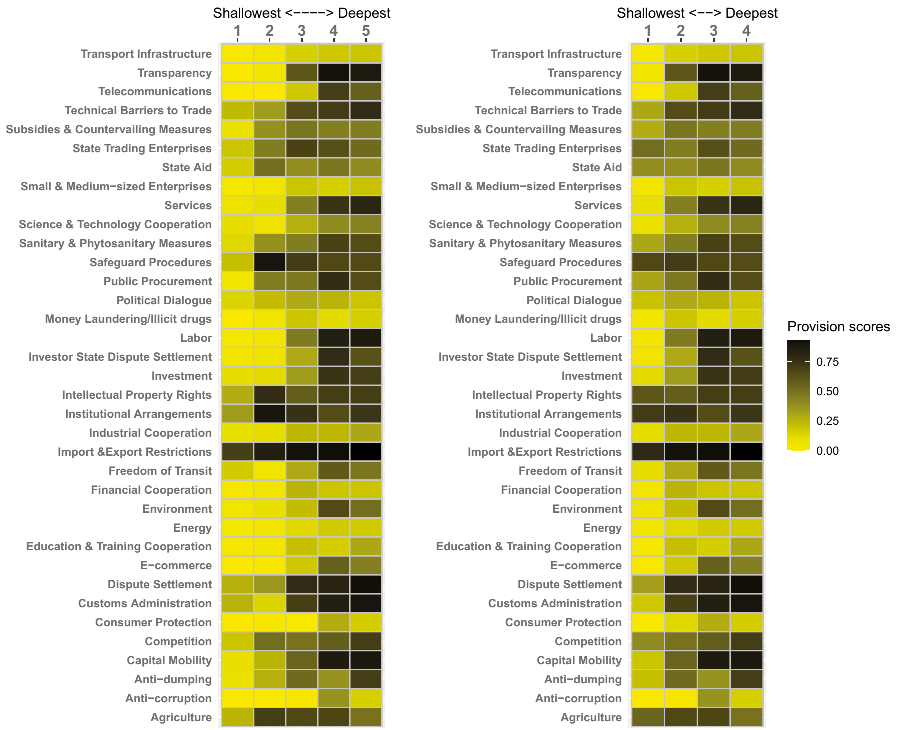

Fig. 4 Heatmap of Provision Scores for Five Clusters and Four Clusters, Respectively, without Removing Extremely Common Phrases/Words

the depth measure in Mattoo et al. (2022), the coefficients estimates in our PPML estimation are roughly the same order of magnitude.

It is interesting to note that our research also enables us to identify the specific provisions that are predominantly present in these highly trade-generating agreements. Table 4 shows the average cluster scores for each provision across all five clusters. For better readability, we present a heatmap for cluster scores in Fig. 4. For consistency with gravity regression results, we have reordered clusters so that number *1* refers to the shallowest cluster and number *5* refers to the deepest cluster. For each provision, the darker the cell, the higher the cluster's performance in that specific provision. The prominent provisions in the *deepest* cluster are antidumping, capital mobility, competition, customs harmonization, dispute settlement mechanism, e-commerce, environment, export and import restrictions, freedom of transit, investment, investor-state dispute settlement, labor, public procurement, sanitary and phytosanitary measures, services, technical barriers to trade, telecommunications, and transparency. We can compare these results to provisions in Breinlich et al. (2021) who find that the provisions that are associated with higher trade volumes include technical barriers to trade, antidumping, trade facilitation, transparency, freedom of transit, subsidies, and competition policies. We find most of these are associated with our deeper agreements.

There are some provisions—such as e-commerce, freedom of transit, and environment laws—for which the cells are not very dark because these provisions are relatively less frequent than the other provisions, and because the scores on these provisions are merely the average across the trade agreements in each cluster. One should interpret these provisions' scores relative to other clusters.

The most comprehensive set of trade agreements remains the most influential in increasing trade flows when we split trade agreements into only four clusters. The third and fourth columns of Table 9 show the gravity trade regression results in the case of four clusters. The FTA coefficient is about the same, 0.632 for OLS and 0.158 for PPML. This result should not be a surprise given the high degree of correlation between the clusters discussed above. The same provisions feature prominently in this cluster as well, as evidenced by high scores on those provisions in Table 5 or the dark squares in the second column of Fig. 4.

The FTA coefficients for the next most comprehensive cluster, "deep," are also quite high for OLS and PPML. The coefficients are 0.624 and 0.137, respectively. Examining the provision scores reveals that the similar set of provisions that was prominently present in the most comprehensive agreements is also dominantly present, albeit with relatively less comprehensiveness and a lower degree of legal enforceability.

One crucial question that remains unanswered is whether the benefits of incorporating the "trade-inducing" provisions extend to an arbitrarily chosen pair of countries. In the context of our study, despite only covering commitments in export and import restrictions why do some of the least comprehensive agreements generate a comparable level of trade flows to some of the clusters that were classified as "moderate," which tends to cover a wide range of policy areas albeit with less comprehensive coverage than the "deeper" and "deepest" agreements. Upon further examination of the agreements in the "shallowest" cluster, it turns out that the majority of these agreements are between transition economies. Some examples include Armenia-Ukraine, the Eurasian Economic Community, Georgia-Ukraine, Kazakhstan-Kyrgyzstan, Russia-Azerbaijan, and Russia-Belarus. If we have agreements between transition economies, it might be the mere establishment of a trade agreement that strongly and positively affects bilateral trade. It may be the case that bilateral trade declined after the Cold War and the trade agreements helped rekindle some of the old trade patterns.

7 Robustness Analysis

We also perform our clustering analysis by removing extremely common and extremely rare words and phrases, as well as overly used words and phrases. In particular, we ignore words and phrases that appear in fewer than one percent of the trade agreements. The idea is to avoid rare words and phrases, such as the names of the parties involved in the trade agreement, that may be driving our clustering results. Similarly, we ignore single words and two-word phrases that appear in more than 99 percent of the trade agreements. The idea is to avoid common words such as *chapter*, *annexes*, *party*, *parties*, and so forth that are quite prevalent in all trade agreements. The optimal number of clusters did not

Table 10 Correlation between the Two Sets of Clusters for Robustness Analysis

	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Cluster 1	**0.45**	−0.23	−0.17	−0.24
Cluster 2	**0.61**	−0.30	−0.24	−0.34
Cluster 3	−0.50	**0.99**	−0.13	−0.18
Cluster 4	−0.39	−0.13	**1.00**	−0.14
Cluster 5	-0.54	-0.18	-0.14	**1.00**

This table shows the correlation matrix between the two sets of clusters. As we move from five clusters to four clusters, cluster 1 and cluster 2 merge into one cluster, while the rest of the clusters do not change as indicated by correlation of 1.00 except for cluster 3

change in this case. The appropriate number of clusters as suggested by the elbow method remains the same, further validating our results.

As in the baseline results, we run the correlation between the two sets of clusters and present the robustness analyses for five and four clusters. Table 10 shows the correlation matrix between the two sets of clusters. The own-cluster correlation is pretty high for all clusters. The cross-cluster correlation is very low, indicating that most trade agreements stay in their "natural" groups as we change the number of clusters.

The gravity analysis of trade flows and the corresponding provisions that tend to have the largest impact on trade flows remain quite stable. The first column of Table 11 shows the gravity trade regression results in the case of five clusters, and Table 12 shows the average cluster scores for each provision. As before, we also present a heatmap for better readability in Fig. 4.

The gravity results indicate that the most comprehensive set of agreements, denoted by *deepest*, have the largest impact on trade flows. The coefficient for this cluster is about

Table 11 Gravity Regressions-Robustness Analysis

Cluster	OLS FE 5 clusters	PPML FE 5 clusters	OLS FE 4 clusters	PPML FE 4 clusters
Shallowest	0.435*** (0.028)	0.128*** (0.014)	0.320*** (0.020)	0.094*** (0.011)
Shallow	0.208*** (0.025)	0.077*** (0.013)	-	-
Moderate	0.351*** (0.026)	0.041*** (0.014)	0.321*** (0.025)	0.048*** (0.013)
Deep	0.622*** (0.028)	0.138*** (0.119)	0.623*** (0.028)	0.137*** (0.012)
Deepest	0.639*** (0.028)	0.162*** (0.012)	0.629*** (0.031)	0.158*** (0.012)
$\bar{R}2$	0.8296	-	0.8296	-
N	611,681	611,681	611,681	611,681

Table reports the coefficients on the gravity regressions with country-pair fixed effects, country-import fixed effects, and country-export fixed effects

***1%, **5%, *10%

Table 12 Provision Scores for Five Clusters for Robustness Analysis

Provisions	Shallowest	Shallow	Moderate	Deep	Deepest
Agriculture	0.25	0.70	0.66	0.67	0.49
Anti-Corruption	0.00	0.00	0.02	0.37	0.15
Anti-Dumping	0.08	0.27	0.52	0.37	0.70
Capital Mobility	0.09	0.25	0.55	0.87	0.89
Competition	0.19	0.51	0.49	0.57	0.70
Consumer Protection	0.00	0.016	0.013	0.28	0.16
Customs Administration	0.26	0.14	0.69	0.85	0.90
Dispute Settlement	0.27	0.36	0.78	0.82	0.92
E-commerce	0.00	0.02	0.18	0.56	0.44
Education and Training Cooperation	0.02	0.04	0.21	0.16	0.30
Energy	0.03	0.06	0.13	0.17	0.17
Environment	0.05	0.09	0.22	0.65	0.51
Financial Cooperation	0.02	0.06	0.25	0.19	0.19
Freedom of Transit	0.17	0.06	0.29	0.59	0.48
Import & Export Restrictions	0.69	0.85	0.91	0.92	0.93
Industrial Cooperation	0.09	0.10	0.24	0.24	0.30
Institutional Arrangements	0.34	0.90	0.75	0.64	0.73
Intellectual Property Rights	0.28	0.77	0.57	0.70	0.70
Investment	0.10	0.12	0.34	0.74	0.70
Investor-State Dispute Settlement	0.04	0.07	0.29	0.78	0.62
Labor	0.03	0.06	0.46	0.84	0.87
Money Laundering/Illicit Drugs	0.00	0.05	0.19	0.11	0.15
Political Dialogue	0.14	0.22	0.29	0.25	0.19
Public Procurement	0.05	0.45	0.47	0.77	0.64
Safeguard Procedures	0.21	0.90	0.71	0.66	0.65
Sanitary and Phytosanitary Measures	0.13	0.38	0.45	0.68	0.64
Science & Technology Cooperation	0.10	0.07	0.27	0.40	0.43
Services	0.07	0.10	0.44	0.74	0.80
Small and Medium-sized Enterprises	0.03	0.05	0.19	0.15	0.20
State Aid	0.16	0.51	0.39	0.48	0.40
State Trading Enterprises	0.19	0.45	0.68	0.63	0.52
Subsidies & Countervailing Measures	0.08	0.38	0.48	0.44	0.45
Technical Barriers to Trade	0.23	0.34	0.64	0.71	0.78
Telecommunications	0.01	0.00	0.18	0.70	0.56
Transparency	0.00	0.06	0.60	0.91	0.88
Transport Infrastructure	0.01	0.04	0.15	0.18	0.19

Table reports the average provision scores across trade agreements within each cluster in the case of 5 clusters for robustness analysis. Low provision scores across all clusters implies that the provision is only prevalent in a few trade agreements

0.639. This indicates that the most comprehensive trade agreements can increase two members' trade flows by up to 90 percent ($e^{0.639} = 1.895$). The prominent provisions in the "deepest" cluster are antidumping, capital mobility, competition laws, customs

Table 13 Provision Scores for Four Clusters for Robustness Analysis

Provisions	Shallowest	Moderate	Deep	Deepest
Agriculture	0.54	0.66	0.67	0.49
Anticorruption	0.00	0.02	0.37	0.15
Antidumping	0.21	0.52	0.37	0.70
Capital mobility	0.19	0.55	0.87	0.89
Competition	0.40	0.49	0.57	0.70
Consumer protection	0.01	0.13	0.28	0.16
Customs administration	0.18	0.69	0.85	0.90
Dispute settlement	0.33	0.78	0.82	0.92
E-commerce	0.01	0.18	0.56	0.44
Education & training cooperation	0.03	0.21	0.16	0.30
Energy	0.05	0.13	0.17	0.17
Environment	0.07	0.22	0.65	0.51
Export & import restrictions	0.79	0.91	0.92	0.93
Financial cooperation	0.05	0.25	0.19	0.19
Freedom of transit	0.10	0.29	0.59	0.48
Industrial cooperation	0.10	0.24	0.24	0.30
Institutional arrangements	0.71	0.75	0.64	0.73
Intellectual property rights	0.60	0.57	0.70	0.70
Investment	0.11	0.34	0.74	0.71
Investor-state dispute settlement	0.06	0.29	0.78	0.62
Labor	0.05	0.46	0.84	0.87
Money laundering/illicit drugs	0.04	0.19	0.11	0.15
Political dialogue	0.20	0.29	0.25	0.19
Public procurement	0.31	0.47	0.77	0.64
Safeguard procedures	0.66	0.71	0.66	0.64
Sanitary & phytosanitary measures	0.30	0.45	0.68	0.64
Science & technology cooperation	0.08	0.27	0.40	0.43
Services	0.09	0.44	0.74	0.80
Small & medium-sized enterprises	0.04	0.19	0.15	0.20
State aid	0.39	0.39	0.48	0.40
State trading enterprises	0.51	0.45	0.63	0.52
Subsidies & countervailing measures	0.28	0.48	0.44	0.45
Technical barriers to trade	0.30	0.64	0.71	0.78
Telecommunications	0.01	0.18	0.70	0.56
Transparency	0.04	0.60	0.91	0.88
Transport infrastructure	0.03	0.15	0.18	0.19

Table reports the average provision scores across trade agreements within each cluster in the case of 4 clusters for robustness analysis. Low provision scores across all clusters implies that the provision is only prevalent in a few trade agreements

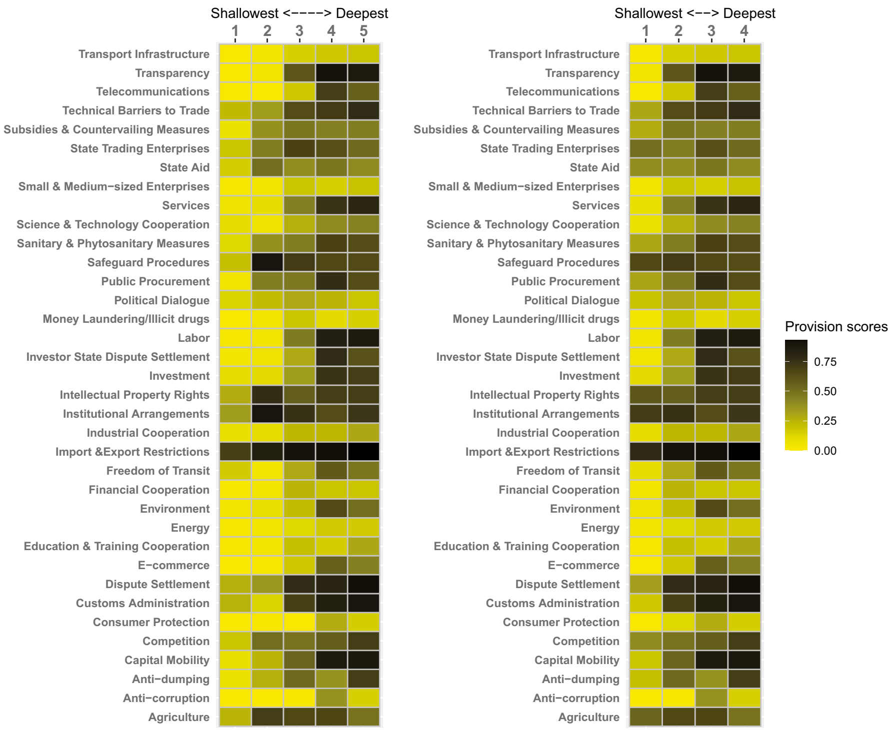

Fig. 5 Heatmap of Provision Scores for Five Clusters and Four Clusters, Respectively, when Removing Extremely Common and Rare Words/Phrases (Robustness Analysis)

harmonization, dispute settlement mechanism, e-commerce, environment, export and import restrictions, freedom of transit, investment, investor-state dispute settlement, labor, public procurement, sanitary and phytosanitary measures, services, technical barriers to trade, telecommunications liberalization, and transparency.

The most comprehensive set of agreements remains the most influential with regard to increasing trade flows as we split trade agreements into only four clusters. The third column of Table 11 shows the gravity trade regression results with four clusters. The FTA coefficient is 0.629. The prominent provisions present in these clusters are the same, as evidenced by high scores on the same set of provisions in Table 13 and Fig. 5.

8 Conclusion and Future Work

In this paper, we capture heterogeneity in free trade agreements using the tools of machine learning. The tools of machine learning allow us to quantify several features of trade agreements, including volume, comprehensiveness, and legal enforceability.

First, we employ unsupervised learning techniques to categorize agreements into four to five clusters. Second, we use supervised learning techniques to analyze the content in each cluster in relation to the coverage of policy areas and the legal enforceability of those provisions. Finally, assuming that the trade flow effects are common across agreements within each cluster, we run the gravity model of trade flows to estimate the differential effects of free trade agreements. We find that more comprehensive agreements result in larger trade creation. In addition, we also identify the provisions that are generally the most successful in driving trade flows. The provisions are antidumping, capital mobility, competition, customs harmonization, dispute settlement mechanism, e-commerce, environmental standards, export and import restrictions, freedom of transit, investment, investor-state dispute settlement, labor, public procurement, sanitary and phytosanitary measures, services, technical barriers to trade, telecommunications, and transparency.

Another question worth exploring is how the content and legal enforceability of a specific provision or a set of provisions affects trade flows in the relevant sector. For example, what aspects of provisions related to automobiles affect the trade flows in automobiles across the member countries? Is one method of rules of origin determination for tariff concessions more restrictive and trade hindering than the other? The answers to such questions may also be obtained by combining a standard empirical trade model with the text analysis of provisions enabled by machine learning. We believe that this stream of research will have far-reaching implications in shaping modern trade institutions.

Appendix A Countries Included in the Gravity Dataset

Albania, Angola, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Brazil, Brunei Darussalam, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Canada, Cape Verde, Central African Republic, Chad, Chile, China, Colombia, Comoros, Congo, Costa Rica, Côte d'Ivoire, Croatia, Cyprus, Czech Republic, Democratic Republic of the Congo, Denmark, Djibouti, Dominica, Dominican Republic, Ecuador, Egypt, Equatorial Guinea, Estonia, Ethiopia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Guinea, Guinea-Bissau, Honduras, Hong Kong SAR (China), Hungary, Iceland, India, Indonesia, Iran, Iraq, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Korea, Kuwait, Kyrgyzstan, Lao People's Democratic Republic, Latvia, Lebanon, Lesotho, Lithuania, Luxembourg, Macao, Macedonia, Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Moldova, Mongolia, Morocco, Mozambique, Namibia, Nepal, Netherlands, New Zealand, Niger, Nigeria, Norway, Oman, Pakistan, Panama, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Romania, Russia, Rwanda, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, São Tomé and Príncipe, Saudi Arabia, Senegal, Sierra Leone, Singapore, Slovakia, Slovenia, South Africa, Spain, Sri Lanka, Sudan, Suriname, Swaziland, Sweden, Switzerland, Syrian Arab

Republic, Tajikistan, Tanzania, Thailand, Togo, Trinidad and Tobago, Tunisia, Turkey, Turkmenistan, Uganda, Ukraine, United Kingdom, United States, Uruguay, Uzbekistan, Venezuela, Vietnam, Yemen, Zambia, and Zimbabwe

Appendix B Trade Agreements by Year of Enforcement

Before 1970: European Economic Community (EEC) (1957), European Community (1958), European Free Trade Agreement (EFTA) (1960), EEC-Turkey (Ankara Agreement) (1963), Southern African Customs Union (1969)

1972: EFTA-EEC (Austria), EFTA-EEC (Norway), EFTA-EEC (Portugal), EFTA-EEC (Sweden), EFTA-EEC (Switzerland)

1973: Caribbean Community (CARICOM), EC-Iceland

1974: EC-Cyprus

1975: EEC-Israel

1981: Gulf Cooperation Council, EEC-Greece

1983: Australia–New Zealand FTA

1985: EEC–Spain, Portugal, USA-Israel

1988: Andean Community (Cartagena Agreement), Canada–US Free Trade Agreement (1988)

1991: Common Market for Eastern and Southern Africa (COMESA)

1992: Association of Southeast Asian Nations (ASEAN), Central European Free Trade Agreement (CEFTA), EC–Czech Republic, EFTA-Slovakia, EFTA-Turkey, European Union Treaty

1993: Armenia-Russia, Czech Republic–Slovakia, EC-Hungary, EFTA-Bulgaria, EFTA–Czech Republic, EFTA-Hungary, EFTA-Israel, EFTA-Poland, EFTA-Romania, EFTA-Slovakia, EU-Poland FTA, Russia-Azerbaijan, Russia-Belarus, Russia-Kazakhstan, Russia-Tajikistan, Russia-Turkmenistan, Russia-Uzbekistan

1994: Baltic Free Trade Agreement–Industrial FTA, CARICOM-Colombia, EC-Bulgaria, European Economic Area, North American Free Trade Agreement, Russia-Georgia, Russia-Kyrgyzstan, Russia-Ukraine

1995: COMESA, EC-Israel, EC-Latvia, EC-Lithuania, EC-Turkey, EFTA-Slovenia, Mercosur (Argentina, Brazil, Paraguay, Uruguay), Mexico-Bolivia, Mexico-Colombia-Venezuela, Mexico–Costa Rica, West African Economic Monetary Union (WAEMU)

1996: Armenia-Kyrgyzstan, Armenia-Moldova, Azerbaijan-Ukraine, Bolivia-Chile, Canada-Chile, Czech Republic–Estonia, Czech Republic–Israel, EC-Morocco, EFTA-Estonia, EFTA-Latvia, Kazakhstan-Kyrgyzstan, Mercosur-Bolivia, Mercosur-Chile, Turkey-Israel, Turkmenistan-Ukraine, Uzbekistan-Ukraine

1997: Armenia-Turkmenistan, Armenia-Ukraine, Canada-Israel, Czech Republic–Latvia, Czech Republic–Lithuania, Czech Republic–Turkey, EFTA-Lithuania, EFTA-Morocco, Estonia-Slovenia, Estonia-Ukraine, Georgia-Azerbaijan, Georgia-Ukraine, Hungary-Israel, Israel–Slovak Republic, Kyrgyzstan-Moldova, Latvia-Slovenia, Lithuania-Slovenia, Macedonia-Slovenia, Poland-Israel, Poland-Lithuania, Slovak Republic–Estonia, Slovak Republic–Latvia, Slovak Republic–Lithuania, Turkey-Hungary

1998: Chile-Mexico, EC-Estonia, EC-Tunisia, India–Sri Lanka, Kyrgyzstan-Ukraine, Mercosur–Andean Community, Pan Arab Free Trade Agreement (PAFTA), Turkey-Bulgaria

1999: Armenia-Georgia, CEFTA-Bulgaria, EC-Slovenia, Egypt-Jordan, Egypt-Morocco, Hungary-Estonia, Israel-Slovenia, Kyrgyzstan-Uzbekistan, Lithuania-Turkey, Poland-Latvia, Turkey-Estonia, Turkey-Macedonia, Turkey-Poland, Turkey–Slovak Republic, SICA (Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama)

2000: Bulgaria-Macedonia, Central African Economic and Monetary Community (CEMAC), EC-Mexico, EC–South Africa FTA, EFTA-Morocco, Georgia-Kazakhstan, Georgia-Turkmenistan, Hungary-Latvia, Hungary-Lithuania, Mexico-Israel, New Zealand–Singapore, WAEMU

2001: Bosnia and Herzegovina–Croatia, East African Community (EAC), EFTA-FYROM, Guatemala-Mexico, Honduras-Mexico, Southern African Development Community, Turkey-Latvia, Turkey-Slovenia

2002: Armenia-Kazakhstan, Bulgaria-Israel, Central America–Dominican Republic, CARICOM–Dominican Republic, Chile–Costa Rica, EC-Croatia, EC-Jordan, EC-Macedonia, EFTA-Croatia, EFTA-Jordan, EFTA-Mexico, Eurasian Economic Community, South African Customs Union, Turkey–Bosnia and Herzegovina, Turkey-Croatia

2010:ASEAN-China, ASEAN-India, ASEAN-Japan, ASEAN–New Zealand–Australia, Canada-Peru, China–Costa Rica, China-Peru, EFTA-Albania, Eurasian Economic Community Customs Union, India-Korea, India-Nepal, India-Thailand, Japan-Vietnam, Peru-Singapore, Switzerland-Japan

2011: Malaysia–New Zealand, Turkey-Jordan

2012: Albania-Iceland, Albania-Norway, D-8 Preferential Trade Agreement, EFTA-Colombia, EFTA-Peru, EU-Korea, Japan-Peru, Korea-EU, Korea-Peru, Malaysia-Chile, Malaysia-India

Appendix C From Text Documents to Numerical Feature Vectors

For either clustering or classification analysis, the text documents must first be converted to a vector of real numbers. We follow a three-step procedure commonly employed in natural language processing literature to transform text documents into numerical feature vectors. The first step involves assigning integer identification for each word or a two-word combination, commonly referred to as tokenization. The trade agreement documents were tokenized using unigram (single word) and bigram counts (two-word phrases). The words for tokenization are defined as sequences of two or more alphabetic characters, excluding stop words, such as pronouns, articles, and prepositions that carry little meaning in differentiating one set of documents from another. We also remove punctuation, numbers, and white spaces. The second step is to count the number of occurrences of these tokens for each document in the collection of documents, commonly referred to as the corpus. The final step is to normalize each document, so it has a feature matrix of fixed size and to weight tokens that occur in the majority of documents with diminishing importance. We use the tf-idf scheme developed by Salton and McGill (1983) to obtain weights for each token.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

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